

Theorems in Lie Algebras

F shall denote an algebraically closed field of characteristic 0 unless the label is starred. L shall always denote a Lie algebra over F . V shall denote a finite dimensional vector space over F .

- 1*. Let $L \subset \mathfrak{gl}(V)$ be a subalgebra consisting of nilpotent endomorphisms, then there exists nonzero $v \in V$ such that $L \cdot v = 0$.

Proof. Induct on $\dim L$. For a subalgebra $K \subset L$ show that $K \subsetneq N_L(K)$ using the image of $\text{ad} : K \rightarrow \mathfrak{gl}(L/K)$ and induction. Choose a maximal subalgebra K of L , then it is an ideal of codim 1. Write $L = K + Fz$ for some $z \in L$. Put $W = \{x \in V \mid L \cdot x = 0\}$, show that it is an L -stable subspace of V . Then an eigenvector of $z|_W$ does the trick.

- 2*. (Engel) L is nilpotent if and only if all elements of L are ad-nilpotent.

Proof. ($\Leftarrow$) Apply the previous lemma to $\text{ad}(L) \subset \mathfrak{gl}(L)$ to get a nonzero $x \in L$ such that $\text{ad}(L) \cdot x = 0$, then $Z(L) \neq 0$. Induct on $\dim L$, to get that $L/Z(L)$ is nilpotent. This shows L nilpotent.

3. (Lie) Let $L \subset \mathfrak{gl}(V)$ be a solvable subalgebra, then there exists nonzero $v \in V$ that is a common eigenvector of L .

Proof. Induct on $\dim L$. Note that $L/[LL]$ is nonzero abelian; choose a ideal of codim 1, then its inverse image K is an ideal of codim 1. Write $L = K + Fz$ for some $z \in L$. Using induction, there is linear $\lambda : K \rightarrow F$ and nonzero $v \in V$ such that $xv = \lambda(x)v$ for all $x \in K$. Put $W = \{v \in V \mid xv = \lambda(x)v \ \forall x \in K\}$, show that W is an L -invariant subspace; an eigenvector of $z|_W$ does the trick. To prove W is L -invariant, it suffices to show that $\lambda[xy] = 0$ for $x \in L, y \in K$. For $w \in W, x \in L$ define $W_k = \{x^i w\}_{i=0}^{k-1}$. Let n be the greatest integer such that W_n is linearly independent. Show that the

matrix of $y \in K$ with respect to the basis W_n is upper triangular with trace $n\lambda(y)$. Observe $0 = \text{tr}_{W_n}([xy]) = n\lambda[xy]$.

4. Let $A \subset B \subset \mathfrak{gl}(V)$ be subspaces. Set $M = \{x \in \mathfrak{gl}(V) \mid [x, B] \subset A\}$. If $x \in M$ is such that $\text{tr}(xy) = 0$ for all $y \in M$, then x is nilpotent.

Proof. Let x be as in the proposition, write $x = x_s + x_n$ using the Jordan-Chevalley decomposition. Pick a basis of V such that $x_s = \text{diag}(a_1, \dots, a_n)$ and x_n is strictly upper triangular. It suffices to show that the (dual of the) subspace E generated by a_i as a vector space over $\mathbb{Q}$ is 0. Let $f \in E^*$ and pick $y \in \mathfrak{gl}(V)$ such that $y = \text{diag}_i(f(a_i))$. Use Lagrange interpolation to find $p, q \in F[T]$ with no constant term such that $\text{ad } y = p(\text{ad } x_s)$ and $\text{ad } x_s = q(\text{ad } x)$. It follows that y maps B into A so $y \in M$. Then $\text{tr}(x_s y) = 0$ whence $\sum_i a_i f(a_i) = 0$. Applying f to this yields $f \equiv 0$.

5. (Cartan's Criterion) Let $L \subset \mathfrak{gl}(V)$ subalgebra such that $\text{tr}(xy) = 0$ for all $x \in L$ and $y \in [LL]$. Then L is solvable.

Proof. It suffices to show $[LL]$ is nilpotent. Apply the above theorem with $A = [LL]$ and $B = L$. Observe that $L \subset M$. Let $[x, y] \in [LL]$ and $z \in L$ then $\text{tr}([xy]z) = \text{tr}(x[yz]) = 0$.

6. The lie algebra L is semisimple if and only if its Killing form is non-degenerate.

Proof. ($\Rightarrow$) Show that $S = \{x \in L \mid \kappa(x, y) = 0 \ \forall y \in L\}$ is solvable using Cartan's Criterion. ($\Leftarrow$) Recall that L is semisimple iff L has no nonzero abelian ideals. Let I be an abelian ideal, it suffices to show that $I \subseteq S = 0$. Let $x \in L, y \in I$ then $\text{ad } y \circ \text{ad } x : L \rightarrow I$ and hence $(\text{ad } y \circ \text{ad } x)^2 = 0$.

7. Let L be semisimple, then there are simple ideals $I_1, \dots, I_k$ such that $L = I_1 \oplus \dots \oplus I_k$. This decomposition is unique.

Proof. For any ideal I of L , define $I^\perp = \{x \in L \mid \kappa(x, y) = 0 \ \forall y \in I\}$. Use Cartan's Criterion to show that $I \cap I^\perp = 0$. Use the rank nullity theorem on the map $L \rightarrow I^*$ where $x \mapsto (y \mapsto \kappa(x, y))$ to get $\dim I + \dim I^\perp = \dim L$ (the non degeneracy of κ is required here). Conclude that $L = I \oplus I^\perp$. To prove the theorem, induct on $\dim L$. Let I be a nonzero minimal ideal, then we get a decomposition for $I^\perp$ which proves the theorem.